Predefining an analysis plan

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

THE CONNECTED WEEK

One complete research-case evidence record.

Preserve the original plan, denominator and visible reasons for later changes.

Keep one habit: Recall → W1–W3/W5 → independent case → project evidence → feedback and delayed return.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Privacy and data governance. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Dataset and benchmark limitations. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Multiple comparisons. Give an example and a mistake that the idea helps you avoid.

Privacy and data governance

> **My remembered idea, example and correction**

Dataset and benchmark limitations

> **My remembered idea, example and correction**

Multiple comparisons

> **My remembered idea, example and correction**

W1 · Explain the current example

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- An analysis plan states outcomes, units, eligibility, exclusions, missing handling, baseline, comparisons and uncertainty methods before primary outcomes are inspected. Preserve version history and later deviations; a label alone does not prove chronology. Main plan includes twenty cases, with four failures later proposed for removal because of appearance. This is outcome-informed.

  Keep actual passes out of twenty; other sixteen outcomes are unknown. Only if all sixteen passed would the original rate be 16/20 = 80%. Separate twelve-case picture excludes unreadable files identified before scoring, not readable model failures. Actual exclusions and pass counts are not supplied. Disclose any necessary revision and retain original analysis rather than presenting new choices as old.

  Prespecification improves auditability but cannot repair poor sampling or invalid assumptions. The course performs a fictional paper audit, not a real preregistration or experiment.

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

> **My reasoning and evidence**

W2 · Trace the worked case

> Keep the supplied inputs, comparison rule and evidence together.

Complete task and evidence

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  Keep actual passes out of twenty; other sixteen outcomes are unknown. Only if all sixteen passed would the original rate be 16/20 = 80%. Separate twelve-case picture excludes unreadable files identified before scoring, not readable model failures. Actual exclusions and pass counts are not supplied. Disclose any necessary revision and retain original analysis rather than presenting new choices as old.

  Prespecification improves auditability but cannot repair poor sampling or invalid assumptions. The course performs a fictional paper audit, not a real preregistration or experiment.

- Plan includes all 20 cases. After results, the team removes four failed cases because they make the graph look worse. Show the intermediate steps and give the conclusion with its scope.

Main record	Supported statement	Unsupported shortcut
Original plan	All twenty cases included	Reconstruct eligibility after seeing outcomes
Four cases fail	Failure is known for those four	Assume every remaining case passed
Remove because graph looks worse	Outcome-informed exclusion violates original plan	Call it a predeclared eligibility rule
Original report	Actual passes / 20; numerator not supplied	Report 16/20 as an observed result
Conditional illustration	If all other sixteen passed, 16/20 = 80%	Treat the if-clause as evidence

Main record	Supported statement	Unsupported shortcut
Necessary revision	Retain original record and disclose revision/reason/timing	Erase old plan or present revised analysis as original

My reasoning and evidence

MAIN LESSON · SECTION 7

W3 · Repair a misconception

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- An analysis plan states outcomes, units, eligibility, exclusions, missing handling, baseline, comparisons and uncertainty methods before primary outcomes are inspected. Preserve version history and later deviations; a label alone does not prove chronology. Main plan includes twenty cases, with four failures later proposed for removal because of appearance. This is outcome-informed.

  Keep actual passes out of twenty; other sixteen outcomes are unknown. Only if all sixteen passed would the original rate be 16/20 = 80%. Separate twelve-case picture excludes unreadable files identified before scoring, not readable model failures. Actual exclusions and pass counts are not supplied. Disclose any necessary revision and retain original analysis rather than presenting new choices as old.

  Prespecification improves auditability but cannot repair poor sampling or invalid assumptions. The course performs a fictional paper audit, not a real preregistration or experiment.

- A report omits this qualification: A prespecified plan can still be flawed; transparency prevents hidden changes, not invalid assumptions.

  Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

> **My reasoning and evidence**

MAIN LESSON · SECTION 9

W5 · Explain the limitation

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- An analysis plan states outcomes, units, eligibility, exclusions, missing handling, baseline, comparisons and uncertainty methods before primary outcomes are inspected. Preserve version history and later deviations; a label alone does not prove chronology. Main plan includes twenty cases, with four failures later proposed for removal because of appearance. This is outcome-informed.

  Keep actual passes out of twenty; other sixteen outcomes are unknown. Only if all sixteen passed would the original rate be 16/20 = 80%. Separate twelve-case picture excludes unreadable files identified before scoring, not readable model failures. Actual exclusions and pass counts are not supplied. Disclose any necessary revision and retain original analysis rather than presenting new choices as old.

  Prespecification improves auditability but cannot repair poor sampling or invalid assumptions. The course performs a fictional paper audit, not a real preregistration or experiment.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

> **My reasoning and evidence**

MAIN LESSON · SECTION 4

Try alone · Selected transfer

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Plan excludes malformed **input**s by a rule fixed beforehand. Three of 15 are malformed; 9 of 12 eligible pass. Report.

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Apply a new case independently

My independent explanation, trace and limits

MAIN LESSON · SECTION 10

Project · Keep the research evidence

> Preserve the question, protocol, observations, analysis and limits.

Complete task and evidence

- Add a labelled predefining an **analysis plan evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of predefining an **analysis plan**.

  Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

MAIN LESSON · SECTION 11

Feedback · Repair and return

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use W2: Trace the worked case for Predefining an **analysis plan**. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use the worked case for Predefining an **analysis plan**. Proposed explanation: "The worked result establishes every future case". Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier W4 response copy

REFERENCE · SECTION 8

Reference · Earlier W4 response copy

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Plan excludes malformed **input**s by a rule fixed beforehand. Three of 15 are malformed; 9 of 12 eligible pass. Report.

My reasoning and evidence

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 12

Extra practice · Pictured case

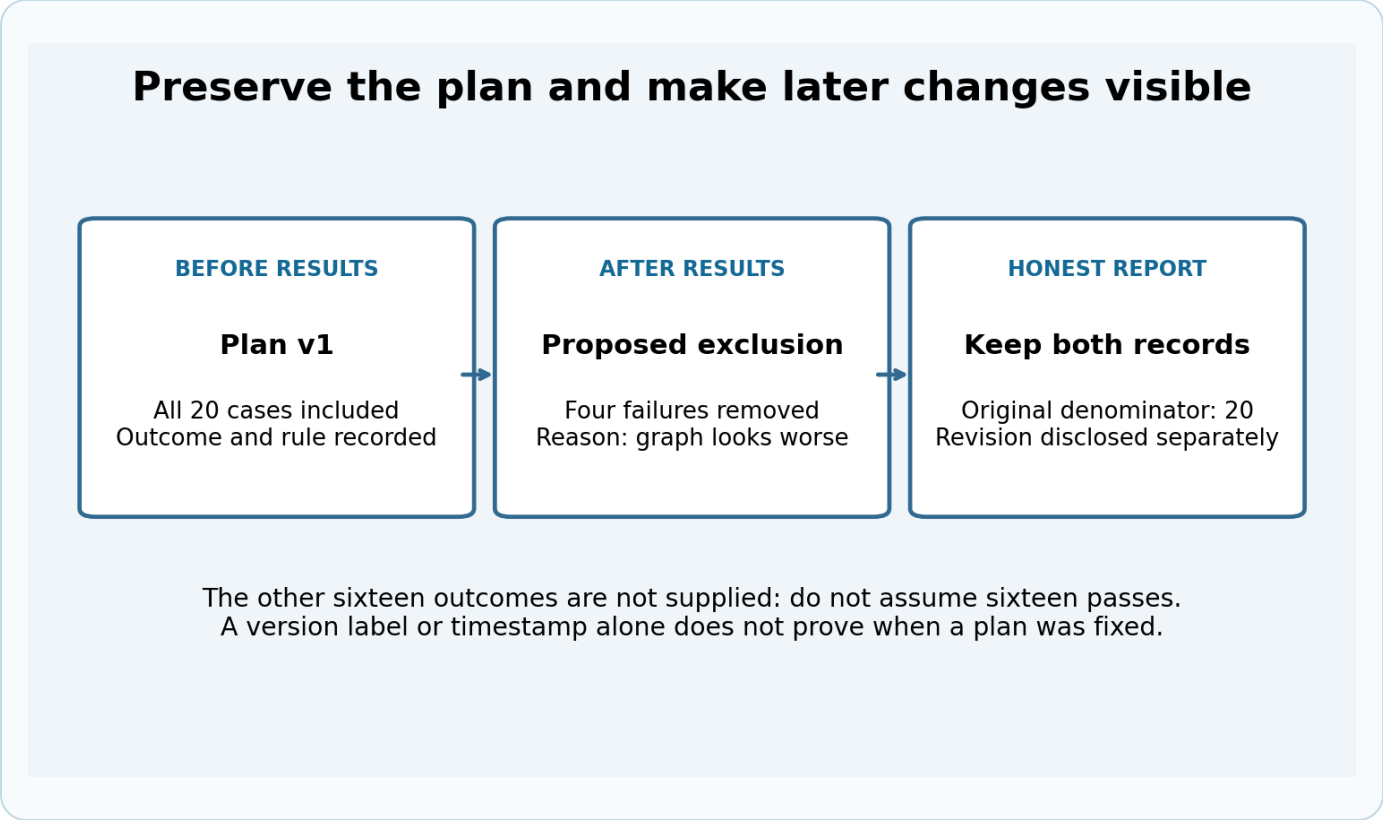
> Keep this task's inputs,rule and response together.

Complete task and evidence

- Freeze decisions before the result. A team records its outcome, exclusions and analysis before evaluating a reserved set.
- Plan: compare correct counts on 12 cases; exclude only unreadable files identified before scoring; report all version results.
- Apply the recorded **exclusion rule**. Can two readable model failures be removed, and how must a necessary later plan change be disclosed?
- Support used: alone / hint / worked example. This record is practice **evidence**.

- Project connection: For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning.

  Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.



AI ACADEMY / AI-7/21

Predefining an analysis plan

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Write the rule first

Record the outcome and justified unreadable-file exclusion.



2. Protect the reserved set

Do not choose exclusions from which cases each model fails.



3. Report deviations

If the plan changes, mark the change and give its reason.

CHANGE ONE THING

Exclude two readable cases because the new model fails them.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

REFERENCE · SECTION 13

Reference · Project milestone

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Project: Reproducible AI Research Study
- Current milestone: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.
- For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ Extra practice · Predict,check and repair

REFERENCE · SECTION 14

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Freeze decisions before the result. A team records its outcome, exclusions and analysis before evaluating a reserved set.
- Which labelled step matches this detail? If the plan changes, mark the change and give its reason.
- 1. Protect the reserved set
- 2. Write the rule first
- 3. Report **deviation**s
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Report **deviation**s If the plan changes, mark the change and give its reason. An **analysis plan** limits flexibility after outcomes are known.

Necessary changes can be transparent, but they should not be silently presented as the original plan.

Changed-case question and optional hint

- Changed-case practice: Exclude two readable cases because the new model fails them. Which conclusion is supported by the supplied facts?
- 1. That outcome-driven exclusion violates the recorded rule.
- 2. This one example guarantees correct results for every future **input**.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: That outcome-driven exclusion violates the recorded rule. Model failure is not the predeclared unreadable-file criterion.

Repair and retain the record

Review Privacy and data governance (ai-7/20). Goal: Explain the **mechanism** and terminology of privacy and data governance. Short practice: Point to the analysis table and the restricted identity key.

Why are the codes pseudonymous rather than guaranteed anonymous, and what changes if the key is published?. Return to this week and explain what you changed.

Review Dataset and benchmark limitations (ai-7/18). Goal: Explain the **mechanism** and terminology of dataset and benchmark limitations. Short practice: Report the indoor benchmark with its denominator and missing condition.

After two correct outdoor cases out of four are added, what slice rate is supported and what remains uncertain?. Return to this week and explain what you changed.

Review Multiple comparisons (ai-7/15). Goal: Explain the **mechanism** and terminology of multiple comparisons. Short practice: Calculate the six-test cutoff and identify which supplied p-values reject.

Why must the team count every planned comparison rather than report only its smallest p-value?. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

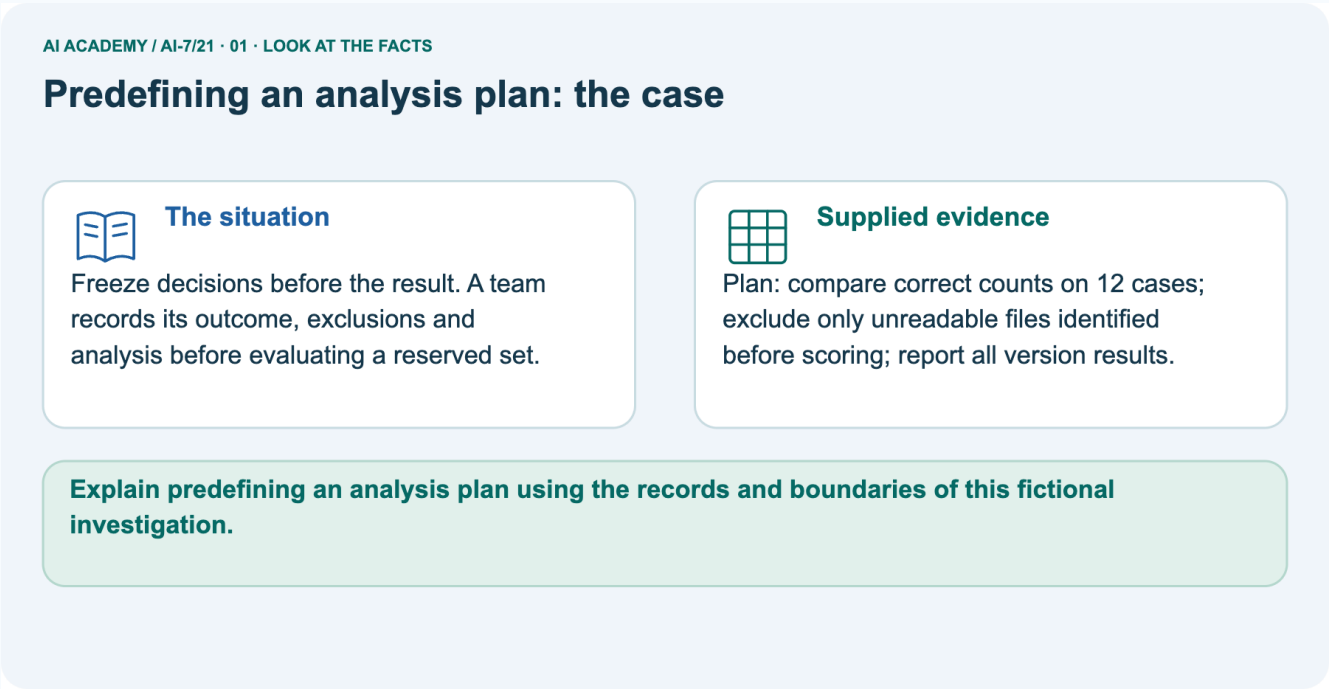
▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 15

Reference · Six-picture response sheets

> Choose the picture that helps explain the current step.

01. Predefining an analysis plan: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

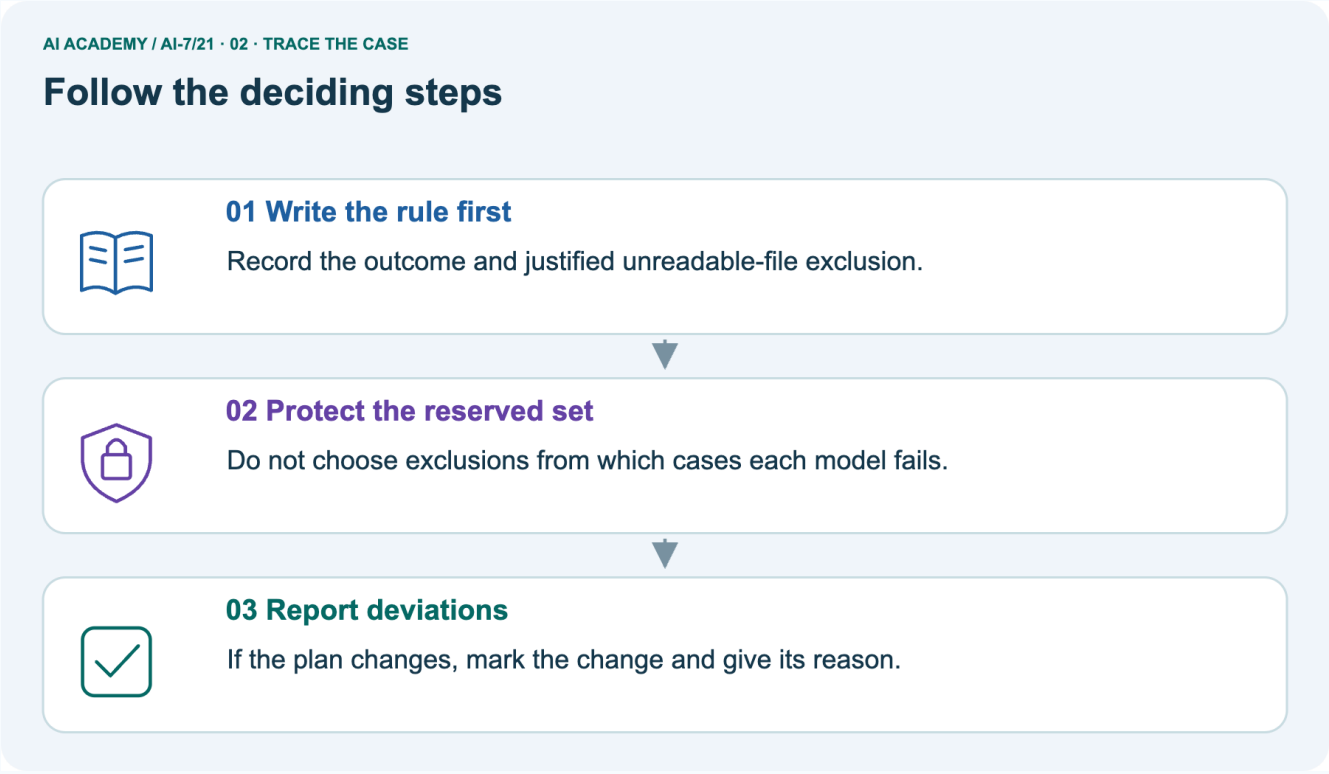
Freeze decisions before the result. A team records its outcome, exclusions and analysis before evaluating a reserved set.

Supplied facts: Plan: compare correct counts on 12 cases; exclude only unreadable files identified before scoring; report all version results.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



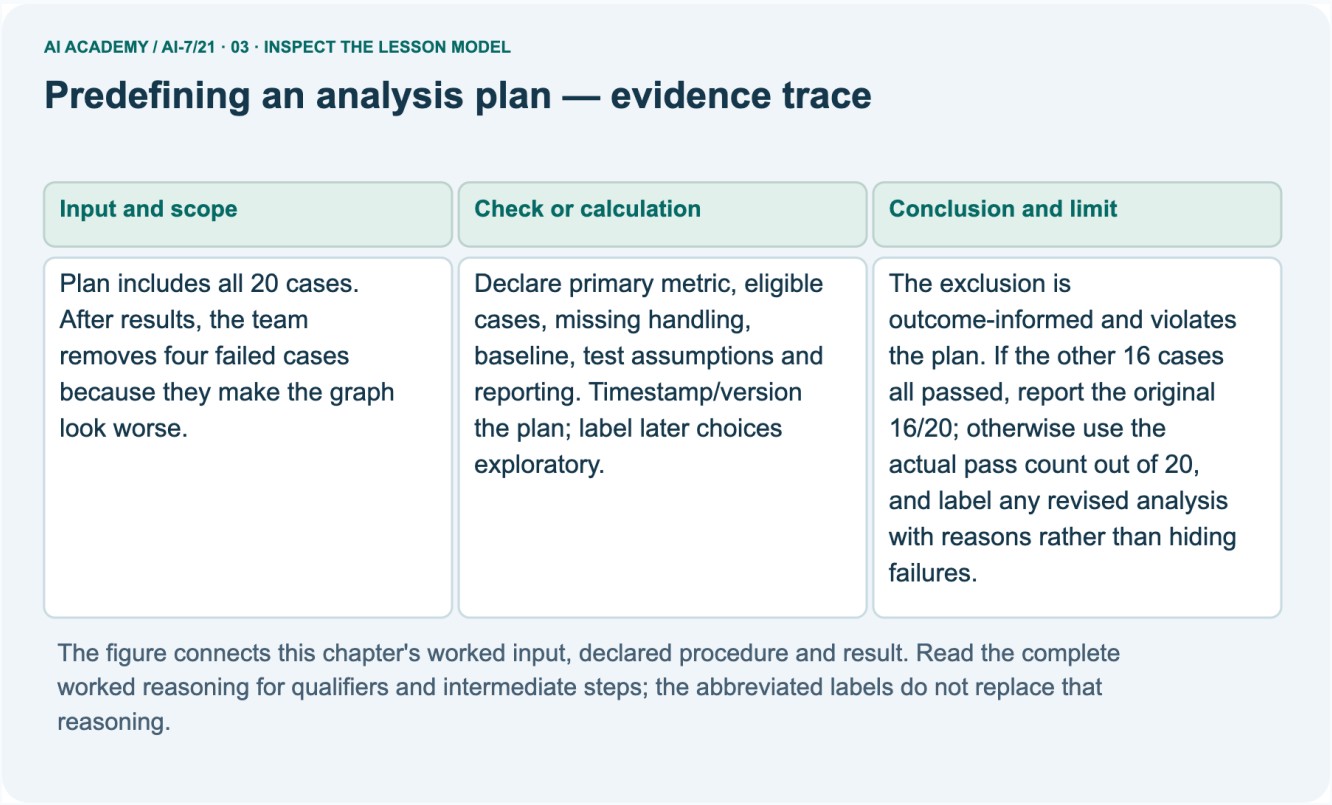
Follow the labelled picture with the matching explanation. Open the vector diagram

Write the rule first: Record the outcome and justified unreadable-file exclusion. → Protect the reserved set: Do not choose exclusions from which cases each model fails. → Report **deviation**s: If the plan changes, mark the change and give its reason.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Predefining an analysis plan — evidence trace



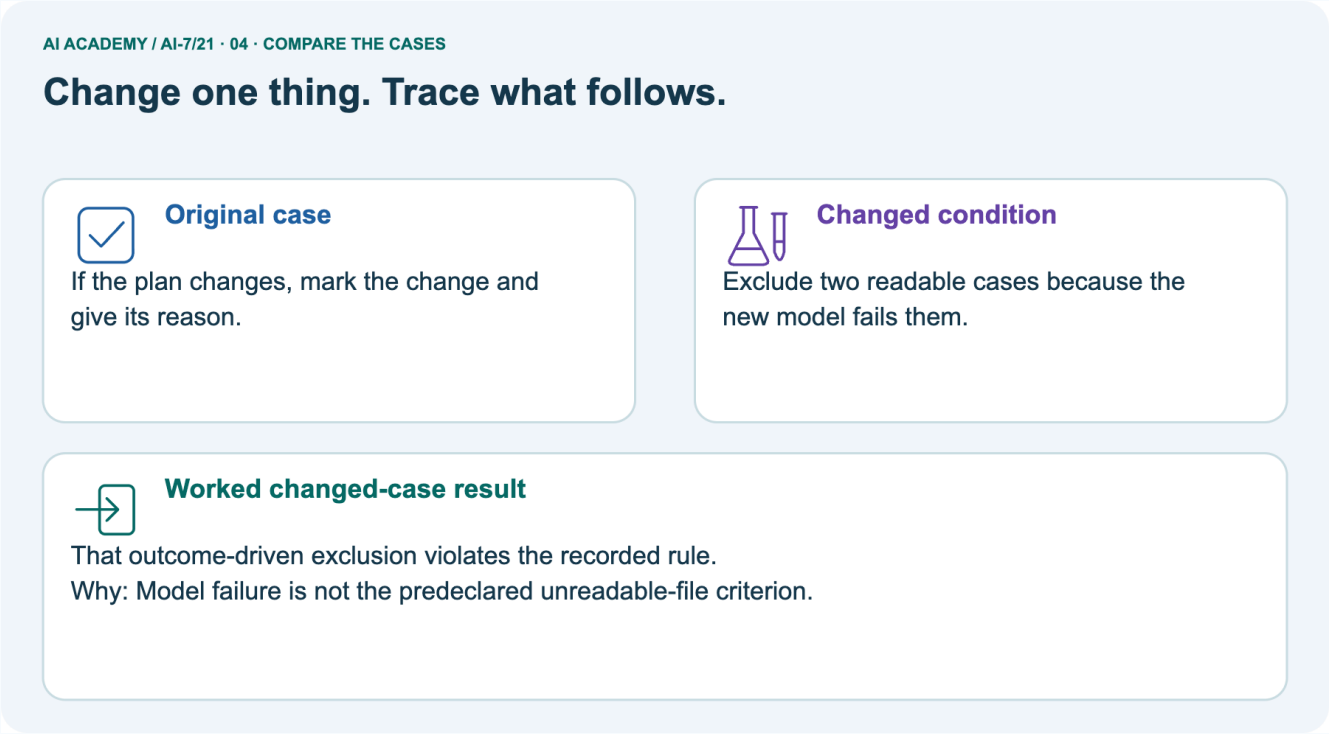
Follow the labelled picture with the matching explanation. Open the vector diagram

Predefining an **analysis plan** — **evidence** trace. Plan includes all 20 cases. After results, the team removes four failed cases because they make the graph look worse. | Declare primary metric, eligible cases, missing handling, baseline, test assumptions and reporting. Timestamp/version the plan; label later choices exploratory. | The exclusion is outcome-informed and violates the plan. If the other 16 cases all passed, report the original 16/20; otherwise use the actual pass count out of 20, and label any revised analysis with reasons rather than hiding failures.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

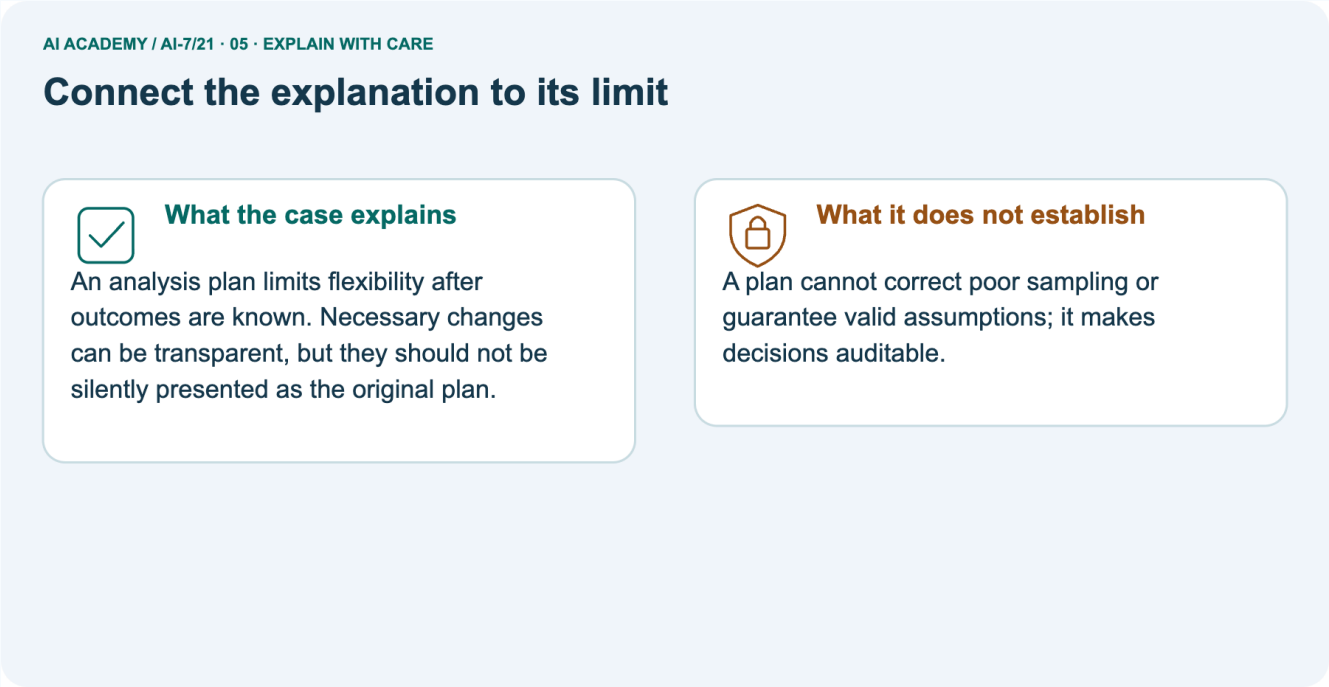
Original case: If the plan changes, mark the change and give its reason. Changed condition: Exclude two readable cases because the new model fails them.

Expected result: That outcome-driven exclusion violates the recorded rule. Reason: Model failure is not the predeclared unreadable-file criterion.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

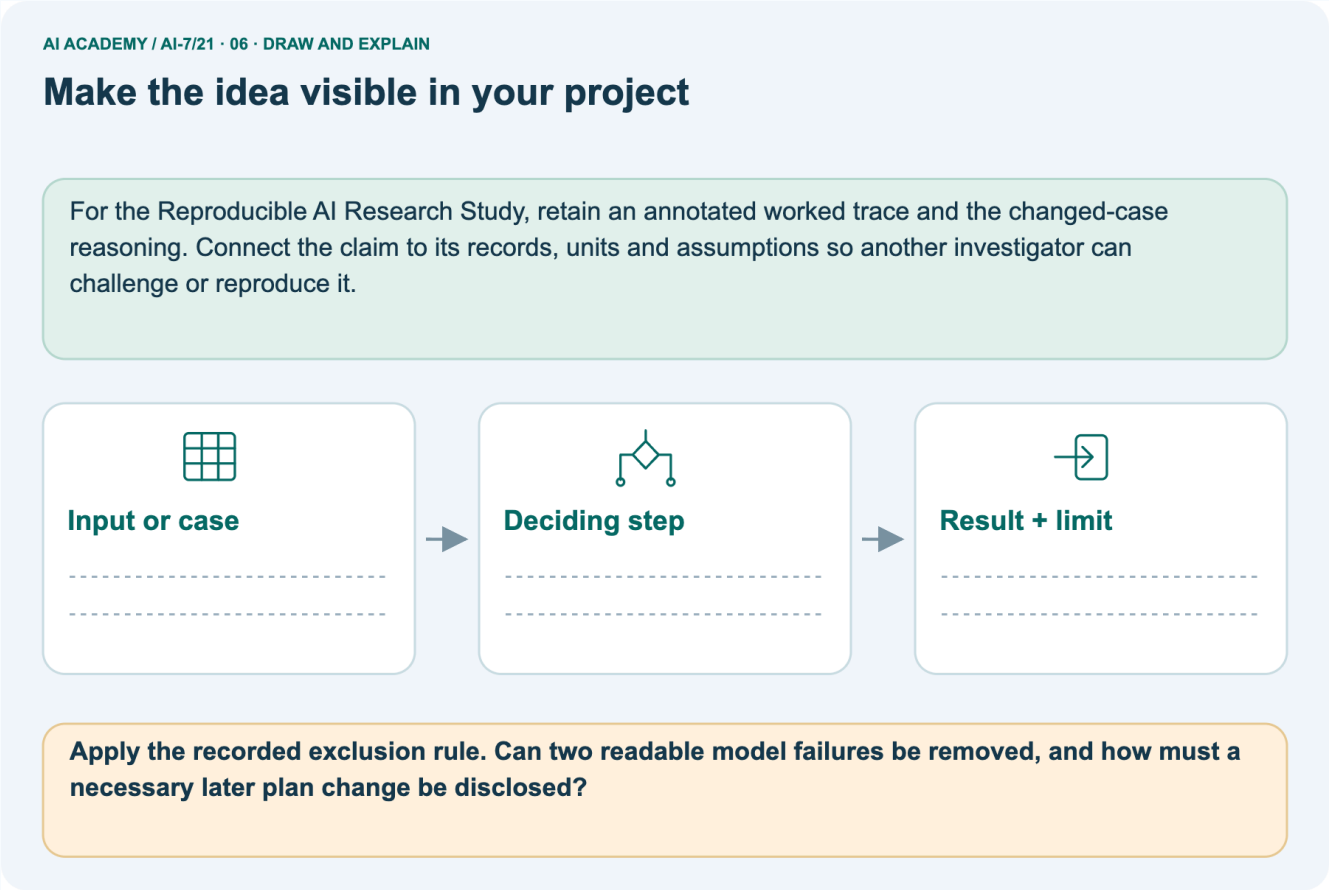
An **analysis plan** limits flexibility after outcomes are known. Necessary changes can be transparent, but they should not be silently presented as the original plan.

Limit: A plan cannot correct poor sampling or guarantee valid assumptions; it makes decisions auditable.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Apply the recorded **exclusion rule**. Can two readable model failures be removed, and how must a necessary later plan change be disclosed?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Predefining an analysis plan: the case

02. Follow the deciding steps

03. Predefining an analysis plan — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.